

MARK WU

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TECHNICAL SKILLS

Languages: Python, TypeScript, Rust, SQL
Web & Backend: React, Tailwind CSS, FastAPI, PostgreSQL, Tauri
ML & Computer Vision: scikit-learn, HRNet
Embedded & CAD: ESP32, Pixhawk, ArduPilot, MAVLink, SolidWorks, Fusion 360
Tools: Git · **Spoken:** English, Mandarin (bilingual)

EDUCATION

University of California, Merced <i>B.S. Mechanical Engineering, Aerospace Engineering emphasis</i>	Merced, CA <i>Expected 2030</i>
Diamond Bar High School <i>AP Computer Science A, AP Calculus BC, AP Physics C: Mechanics, PLTW</i>	Diamond Bar, CA <i>June 2026</i>

EXPERIENCE

FRC Team Sprocket 3473 <i>CAD Subteam; Scouting Software Subteam Founder (Alumnus)</i>	Diamond Bar, CA <i>2024 – 2026</i>
– Founded the scouting software subteam after the legacy app became a competitive liability; served as de facto technical lead for the 5-person team, directing feature decisions and mentoring 3 contributing developers while personally writing the majority of the codebase – Secured a dedicated subteam sponsor (HumanSignal) and managed the infrastructure budget across Neon, Fly.io, and domain hosting – Deployed live at 3 FRC competitions, including both 2026 CA district events, capturing 411 robot-match records across 73 teams – Graduated as an alumnus of the team; still contribute code periodically, taking on harder features, and continue pushing the subteam to grow its membership – Team awards during tenure: Regional FIRST Impact (2025), Engineering Inspiration (2026), Worlds (2024, 2025)	
Team Infernope <i>Combat Robotics, Design and Fabrication</i>	ShenZhen, CN <i>2020 – 2024</i>
– Designed and built twelve combat robots over three years (chassis, drivetrain, weapons, in-match repair); placed 1st in the 2024 end-of-year tournament	
SoCal FTC Qualifiers <i>FIRST Technical Advisor (Volunteer)</i>	Diamond Bar, CA <i>January 2026</i>
– Diagnosed field hardware, software, and robot control issues during match operations at two qualifier events	

PROJECTS

SprocketStats <i>TypeScript, React, Python, FastAPI, PostgreSQL, scikit-learn, HRNet</i>	2025 – Present
– Architected and personally wrote most of a full-stack scouting PWA: React/Tailwind frontend, FastAPI backend, PostgreSQL database – Built a match win prediction pipeline (Random Forest, K-Means) scoring ~87% across ~90 late-qualification and playoff matches, statistically indistinguishable from the Statbotics EPA baseline at 95% confidence in a proportions test – Developing a real-time robot detection and pose tracking pipeline (HRNet) for alliance tracking; designing a multi-target multi-camera tracking (MTMCT) architecture for v3	
Caelifer, Coaxial EDF VTOL <i>SolidWorks, ESP32</i>	2026 – Present
– Designing a tube-launchable VTOL in a missile form factor, targeting multirotor-class hover and maneuverability with improved cruise efficiency and no chemical propulsion – Selected a coaxial contra-rotating EDF stack to cancel both rotor torque and exhaust swirl, packaging the vehicle into a 100 mm finless tubular outer mold line with no external control surfaces – Designed pitch and yaw control around servo-actuated exhaust deflector vanes; the near-axial efflux from the contra-rotating stack keeps vane authority predictable and decoupled – Completed full custom airframe CAD in SolidWorks and ran Flow Simulation studies across the airframe – Bench-testing the propulsion and avionics stack ahead of first hover test: coaxial 70 mm EDF pair on a 6S pack, 180 A peak draw (~4.0 kW)	
Project Aetherius, Fixed-Wing UAV <i>Tauri, TypeScript, Rust, MAVLink, ArduPilot, Pixhawk 6X</i>	2025 – Present
– Building a custom Tauri ground control station (3rd revision, ongoing) with MAVLink telemetry and real-time flight data visualization – Implementing an ArduPilot communication layer over Tauri IPC for parameter access, telemetry, and command execution; validated in SITL for param read/write, pre-arm checks, telemetry streaming, arming, and takeoff, with full mission-sequence testing still in progress – Built a 2m twin-boom airframe (4th revision, 2025–2026) with avionics installed; currently in ground and bench testing, has not yet flown	